Week 6 : Review Details

As The E-commerce company sells multiple product. the **Review table** plays a vital role in capturing customer feedback and ratings for each item. It ensures that every product is linked to genuine customer experiences, allowing the business to evaluate performance, identify top-rated products, and improve overall service quality.

1.Create Review Table

```
SQL> CREATE TABLE Review(Review_ID VARCHAR2(20),
  2  Customer_ID NUMBER(38),
  3  Product_ID VARCHAR2(10),
  4  Rating NUMBER(1),
  5  Review_Test VARCHAR2(60),
  6  Review_Date DATE,
  7  Customer_Name VARCHAR2(25));

Table created.
```

Relationships

- **One-to-Many (1:M):**
    - One **Customer** can write many **Reviews**.
    - One **Product** can have multiple **Reviews**.
- **One-to-One (1:1):**
    - Each **Payment** is linked to exactly one **Order**.
- **Many-to-Many (M:N):**
    - Customers and Products are connected through **Orders** (a customer can buy many products, and a product can be bought by many customers).

Constraints

- **Primary Key (PK):** Ensures each record is unique
- **Foreign Key (FK):** Maintains referential integrity between tables.

1.we add primary key to Order details Table

```
SQL> ALTER TABLE Review
  2  ADD CONSTRAINT FK_Customer
  3  FOREIGN KEY (Customer_ID)
  4  REFERENCES Customer(Customer_ID);

Table altered.
```

2.we added check constrain to check the rating

```
SQL> ALTER TABLE Review ADD CONSTRAINT CHK_RATE CHECK(Rating>=1 AND Rating<=5);

Table altered.
```

Insert values in Review Table

```
SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Product_ID, Rating, Review_Test, Review_Date, Customer_Name)
  3  VALUES
  4  ('REV001', 101, 'P00004', 5, 'Excellent product, highly recommended!', DATE'2026-10-11', 'Sweety');

1 row created.

SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Product_ID, Rating, Review_Test, Review_Date, Customer_Name)
  3  VALUES
  4  ('REV002', 102, 'P00001', 4,'Elegant ceramic bowls, perfect for serving.',DATE'2026-09-03','Arun');

1 row created.

SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Product_ID, Rating, Review_Test, Review_Date, Customer_Name)
  3  VALUES
  4  ('REV003', 103, 'P00002', 3,'Nice look but handle feels fragile.',DATE'2026-09-20','Priya');

1 row created.

SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Product_ID, Rating, Review_Test, Review_Date, Customer_Name)
  3  VALUES
  4  ('REV004', 104, 'P00006', 5, 'Loved it! Worth every penny.',DATE'2026-09-21','Kumar');

1 row created.

SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Product_ID, Rating, Review_Test, Review_Date, Customer_Name)
  3  VALUES
  4  ('REV005', 105, 'P00003', 5, 'Perfect size for my indoor plants',DATE'2026-10-12', 'Meena');

1 row created.

SQL> INSERT INTO Review
  2  (Review_ID, Customer_ID, Product_ID, Rating, Review_Test, Review_Date, Customer_Name)
  3  VALUES
  4  ('REV006', 106, 'P00005', 4, 'Nice wall hanging I love it.', DATE'2026-10-05', 'Ravi');

1 row created.
```

Result

```
SQL> SELECT * FROM REVIEW;

REVIEW_ID            CUSTOMER_ID PRODUCT_ID     RATING
-------------------- ----------- ---------- ----------
REVIEW_TEST                                                  REVIEW_DA
------------------------------------------------------------ ---------
CUSTOMER_NAME
------------------------
REV001                       101 P00004              5
Excellent product, highly recommended!                       11-OCT-26
Sweety

REV002                       102 P00001              4
Elegant ceramic bowls, perfect for serving.                  03-SEP-26
Arun

REVIEW_ID            CUSTOMER_ID PRODUCT_ID     RATING
-------------------- ----------- ---------- ----------
REVIEW_TEST                                                  REVIEW_DA
------------------------------------------------------------ ---------
CUSTOMER_NAME
------------------------

REV003                       103 P00002              3
Nice look but handle feels fragile.                          20-SEP-26
Priya

REV004                       104 P00006              5
Loved it! Worth every penny.                                 21-SEP-26

REVIEW_ID            CUSTOMER_ID PRODUCT_ID     RATING
-------------------- ----------- ---------- ----------
REVIEW_TEST                                                  REVIEW_DA
------------------------------------------------------------ ---------
CUSTOMER_NAME
------------------------
Kumar

REV005                       105 P00003              5
Perfect size for my indoor plants                            12-OCT-26
Meena

REV006                       106 P00005              4

REVIEW_ID            CUSTOMER_ID PRODUCT_ID     RATING
-------------------- ----------- ---------- ----------
REVIEW_TEST                                                  REVIEW_DA
------------------------------------------------------------ ---------
CUSTOMER_NAME
------------------------
Nice wall hanging I love it.                                 05-OCT-26
Ravi


6 rows selected.
```

Report 1 : Customer feedback and Rating

```
SQL> SELECT (c.Customer_Name) AS NAME,
  2  R.Review_ID,R.Product_ID,R.Rating,R.Review_Test
  3  FROM Customer c JOIN Review R
  4  ON c.Customer_ID = R.Customer_ID;

NAME                      REVIEW_ID            PRODUCT_ID     RATING
------------------------- -------------------- ---------- ----------
REVIEW_TEST
------------------------------------------------------------
Sweety                    REV001               P00004              5
Excellent product, highly recommended!

Arun                      REV002               P00001              4
Elegant ceramic bowls, perfect for serving.

Priya                     REV003               P00002              3
Nice look but handle feels fragile.


NAME                      REVIEW_ID            PRODUCT_ID     RATING
------------------------- -------------------- ---------- ----------
REVIEW_TEST
------------------------------------------------------------
Kumar                     REV004               P00006              5
Loved it! Worth every penny.

Meena                     REV005               P00003              5
Perfect size for my indoor plants

Ravi                      REV006               P00005              4
Nice wall hanging I love it.


6 rows selected.
```

Display a product how many time a product seller

```
SQL> SELECT * FROM Review WHERE Product_ID = 'P00006';

REVIEW_ID            CUSTOMER_ID PRODUCT_ID     RATING
-------------------- ----------- ---------- ----------
REVIEW_TEST                                                  REVIEW_DA
------------------------------------------------------------ ---------
CUSTOMER_NAME
-------------------------
REV004                        104 P00006              5
Loved it! Worth every penny.                                 21-SEP-26
Kumar
```

Report 2:Product that has maximim reviews

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      COUNT(R.Review_ID) AS Review_Count
  5  FROM Product P
  6  JOIN Review R
  7      ON P.Product_ID = R.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Review_Count DESC;

PRODUCT_ID PRODUCT_NAME    REVIEW_COUNT
---------- --------------- ------------
P00004     Flower_Vase                1
P00001     Ceramic bowl               1
P00005     Wall hanging               1
P00006     Ceramic Plate              1
P00003     Plant Pot                  1
P00002     Coffee Cup                 1

6 rows selected.
```

Report 3:Display recent feedback

```
SQL> SELECT
  2      C.Customer_Name AS "Customer Name",
  3      R.Review_ID,
  4      R.Product_ID,
  5      R.Rating,
  6      R.Review_Test
  7  FROM Customer C
  8  JOIN Review R
  9  ON C.Customer_ID = R.Customer_ID;

Customer Name             REVIEW_ID            PRODUCT_ID     RATING
------------------------- -------------------- ---------- ----------
REVIEW_TEST
--------------------------------------------------------------------
Sweety                    REV001               P00004              5
Excellent product, highly recommended!

Arun                      REV002               P00001              4
Elegant ceramic bowls, perfect for serving.

Priya                     REV003               P00002              3
Nice look but handle feels fragile.


Customer Name             REVIEW_ID            PRODUCT_ID     RATING
------------------------- -------------------- ---------- ----------
REVIEW_TEST
--------------------------------------------------------------------
Kumar                     REV004               P00006              5
Loved it! Worth every penny.

Meena                     REV005               P00003              5
Perfect size for my indoor plants

Ravi                      REV006               P00005              4
Nice wall hanging I love it.


6 rows selected.
```

Report 4: Highest rated products

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      ROUND(AVG(R.Rating), 2) AS Average_Rating
  5  FROM Product P
  6  JOIN Review R
  7      ON P.Product_ID = R.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  HAVING AVG(R.Rating) = (
 10      SELECT MAX(Avg_Rating)
 11      FROM (
 12          SELECT AVG(Rating) AS Avg_Rating
 13          FROM Review
 14          GROUP BY Product_ID
 15      )
 16  );

PRODUCT_ID PRODUCT_NAME    AVERAGE_RATING
---------- --------------- --------------
P00004     Flower_Vase                 5
P00006     Ceramic Plate               5
P00003     Plant Pot                   5
```